Singapore School Kinderland

Secondary 1 (Sec 1) — Class Test 1 (Term 2) — TEACHER ANSWER KEY

Subject: Computer Science | Chapter 2.2: Methods of Error Detection (TB Pg 54–62) | Max Marks: 25

Candidate Name: _____	**Class / Sec: Sec 1**	**Roll No:** _____
Date: _____	**Time Allowed: 45 Mins**	**Marks Obtained:** MARKING SCHEME

INSTRUCTIONS: Answer ALL questions in the spaces provided. Show all step-by-step calculations and reasoning clearly. No calculators or external electronic devices are permitted.

Question 1: Parity Bit Calculation & Principles *[6 Marks]*

(a) Complete the table by providing the required Parity Bit (most significant bit) and the transmitted byte: *[3]*

Data Byte	7-bit Original Data Payload	Parity System	Parity Bit (0 or 1)	Complete 8-bit Transmitted Byte
Byte A	1 0 1 1 0 1 0	Even Parity	**0 (count=4)**	**0 1 0 1 1 0 1 0**
Byte B	0 1 1 0 1 0 0	Odd Parity	**0 (count=3)**	**0 0 1 1 0 1 0 0**
Byte C	1 1 1 1 0 1 1	Even Parity	**0 (count=6)**	**0 1 1 1 1 0 1 1**

(b) During transmission of Byte A, electromagnetic noise caused two bits to flip simultaneously. Explain why a standard parity check will fail to detect this transmission error: *[3]*

Model Answer [3 Marks]: A parity check only counts whether the total number of 1-bits is even or odd [1]. If two bits flip simultaneously (e.g. two 0s flip to 1s, or two 1s flip to 0s, or one 0 flips to 1 and one 1 flips to 0) [1], the total count of 1-bits still retains the same even/odd parity status [1]. Therefore, the parity rule is satisfied and the system cannot detect the corruption.

Question 2: Parity Block Matrix Error Pinpointing *[6 Marks]*

A transmission block of four bytes was sent using **Even Parity**. A vertical parity byte is included at the bottom:

Row Identifier	Bit 7 (Parity)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Row Status (OK / Error)
Byte 1	0	1	0	1	1	0	1	0	OK (4)
Byte 2	1	1	1	0	0	1	0	1	OK (5)
Byte 3	0	0	1	1	1	0	1	1	**ERROR (5 ones)**
Byte 4	1	1	0	1	0	0	1	0	OK (4)
Parity Byte	**0**	**1**	**0**	**1**	**0**	**1**	**1**	**0**	

(a) Complete the "Row Status" column for each row in the table above. *[2]*

(b) Locate the exact corrupted bit: Corrupted Row: **Byte 3** | Corrupted Column: **Bit 3** *[2]*

(c) Erroneous bit value: **1** | Corrected bit value: **0** *[2]*

Question 3: Checksums & Echo Checks Comparison ***[6 Marks]***

(a) Describe how a checksum is used to verify that a transmitted data block has arrived without corruption. ***[3]***

(b) In an Echo Check, received data is sent back to the sender for verification. Explain TWO major drawbacks of this method: ***[3]***

Drawback 1:

Drawback 2:

Question 4: Check Digits & Human Entry Errors ***[4 Marks]***

(a) Identify TWO types of human data entry errors that Check Digits detect when scanning barcodes: ***[2]***

(b) The first 12 digits of an ISBN-13 barcode are: `9 7 8 0 1 9 8 5 3 4 0 3`.

Calculate the 13th Check Digit using alternating weights of 1 and 3. Show your complete calculation: ***[2]***

Question 5: Automatic Repeat Requests (ARQ) Protocol ***[3 Marks]***

A reliable network protocol uses ARQ for data transfer between Server A and Client B.

(a) Define the purpose of a **Positive Acknowledgement (ACK)** message: ***[1]***

(b) State TWO distinct events that will cause Server A to automatically re-transmit a packet: ***[2]***

Event 1:

Event 2: